USMAN SHAHBAZ

Senior Software Engineer | Lead AI & Mobile Application Engineer

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PROFESSIONAL SUMMARY

Accomplished **Senior Software Engineer & Lead AI Application Engineer** with **6+ years of hands-on experience** architecting, developing, and deploying high-performance applications across **iOS, Android, macOS, Windows, and Web**. Recognized authority in the **Flutter & Dart ecosystem** (Flutter 3.x, Dart 3.x), **Kotlin Multiplatform (KMP)**, and **On-Device & Cloud Generative AI** (Google MediaPipe Pose Biomechanics, OpenAI/ChatGPT APIs, Whisper Speech-to-Text, LangChain). Proven track record leading enterprise application architecture, deploying automated Fastlane CI/CD pipelines, and driving scalable growth via A/B testing and event analytics across systems serving 100K+ global users.

CORE TECHNICAL COMPETENCIES

Languages:	Dart 3.x, Python, Swift (iOS), Kotlin (Android), JavaScript/TypeScript, SQL
Platforms & Cross-Platform:	Flutter (iOS, Android, macOS, Windows, Web), Kotlin Multiplatform (KMP), Jetpack Compose, Native iOS (Swift/Xcode), Android Studio
AI, ML & Computer Vision:	Google MediaPipe (Pose Detection & Biomechanical Angles), OpenAI/ChatGPT API, Whisper (STT), LangChain, ML Kit, Prompt Engineering, Vector DBs, AI Agents
Architecture & State:	Clean Architecture, Riverpod, BLoC, Provider, GetX, SOLID Principles, MVVM, Micro-frontends
Backend, APIs & Cloud:	FastAPI, RESTful APIs, GraphQL, WebSockets (SignalR), Supabase, Firebase (Auth, Firestore, Cloud Functions, Cloud Messaging, Crashlytics)
Local Storage & Offline:	Drift, Isar, SQLite, Hive, SharedPreferences, Offline-First Sync Strategies
Analytics & Monetization:	PostHog, Amplitude (Experimentation & A/B Testing), Customer.io, Intercom, Mixpanel, Sentry, Stripe, In-App Purchases (RevenueCat, StoreKit, Google Play Billing)
DevOps, CI/CD & Testing:	Fastlane (Automated iOS/Android deployments), GitHub Actions, Unit Testing, Widget Testing, Integration Testing, Git

PROFESSIONAL EXPERIENCE

Senior Software Engineer – Nextbridge
12/2023 – Present | Lahore, Pakistan

- Lead technical architecture and multi-platform application development for high-visibility commercial and real estate enterprise clients.
- Architected comprehensive event analytics and A/B testing infrastructure utilizing **Amplitude** and **PostHog**, tracking 20+ user conversion funnels to drive data-informed product iterations and increase user engagement.
- Built and automated complete **CI/CD delivery pipelines via Fastlane**, orchestrating seamless test distribution and multi-tenant store releases to Apple App Store Connect and Google Play Console, reducing deployment turnaround times by **60%**.
- Integrated live in-app user communication channels using **Intercom Messenger and Customer Support SDKs**, improving user feedback collection and support ticket resolution rates.
- Established strict code quality guidelines, modular Clean Architecture conventions, and comprehensive unit/widget testing suites across team codebases.

Senior Software Engineer – RootPointers
02/2021 – 11/2023 | Lahore, Pakistan

- Engineered full-lifecycle multi-platform applications from discovery through store deployment across health, social networking, and enterprise domains.
- Implemented high-concurrency real-time features including **Agora Audio/Video calling**, **WebSocket** synchronizations, and **Firestore** 1-on-1 and group messaging systems.
- Engineered resilient payment gateways using **Stripe Payment Intents** and multi-tiered **In-App Purchases (iOS StoreKit & Google Play Billing)**, ensuring PCI compliance and robust receipt validation.
- Engineered offline-first data persistence layers with **SQLite, Drift, and Hive**, enabling responsive local querying and background synchronization upon network reconnection.
- Mentored a squad of junior developers, conducted code reviews, and championed modern reactive state management patterns (Riverpod, BLoC).

Software Developer (Flutter) – SJ Solutionz
09/2020 – 02/2021 | Lahore, Pakistan

- Translated complex Figma design systems into pixel-perfect, responsive mobile layouts with silky 60fps Flutter animations.
- Integrated RESTful APIs, token-based authentication workflows, and custom interceptors for resilient error handling and network status monitoring.
- Configured cross-platform push notifications, deep links, and device hardware integrations (camera, geolocation, sensors).

FEATURED AI & APPLICATION PROJECTS

TruMotion – Real-Time AI Exercise & Biomechanics Analyzer Kotlin Multiplatform (KMP) | Computer Vision / AI

Stack: Kotlin Multiplatform, Google MediaPipe, Computer Vision, CameraX, Vector Geometry, Live Pose Estimation

- Architected an on-device AI fitness analyzer using **Google MediaPipe Pose Detection** on high-framerate live camera video streams.
- Engineered geometric vector algorithms to calculate real-time anatomical joint angles, body alignment vectors, and repetition counts, delivering instant biomechanical visual feedback without cloud latency.
- Designed a unified **Kotlin Multiplatform (KMP)** shared business and inference pipeline, maximizing native performance across platforms while cutting code duplication by **50%**.

Wakefully: AI Dream Decoder & Mindfulness Platform [App Store] [Google Play] Generative AI / Mobile App

Stack: Flutter, Dart, OpenAI ChatGPT API, Whisper (Speech-to-Text), Customer.io, In-App Purchases, .NET

- Developed core mobile AI flows including **Dream Decoder™**, **Dream Rescript**, and **Dream Manifester** utilizing OpenAI LLM prompt architectures.
- Integrated real-time **Speech-to-Text (Whisper/STT)** dictation to allow frictionless spoken dream journaling and instant emotion classification.
- Integrated **Customer.io SDK** for lifecycle event telemetry, behavioral broadcast triggers, and tailored subscriber journeys, boosting user retention.
- Architected recurring subscription tiers and premium paywalls using In-App Purchases, driving app monetization across iOS and Android.

Betterhomes Engage – Real Estate Enterprise Platform [App Store] [Google Play] Commercial Enterprise App

Stack: Flutter, Dart, Firebase, Amplitude, PostHog, Fastlane, Intercom, Node.js, Google Calendar API

- Engineered flagship real estate agent application managing **8,500+ UAE property listings**, inventory exploration, and lead pools.
- Integrated bi-directional **Google Calendar sync** for seamless property viewing scheduling directly within the agent workflow.
- Implemented Amplitude analytics and PostHog feature flags for dynamic A/B test rollout of new listing interaction patterns.

Shapes – Fitness Center & Staff Facility Platform Enterprise Health & Fitness

Stack: Flutter, Dart, Firebase, REST APIs, QR Code Check-in, WebSockets, State Management

- Engineered a centralized management application for fitness centers, personal trainers, and gym facility administration.
- Implemented member check-in validation via QR scanning, real-time schedule rosters, personal trainer booking, and class capacity monitors.
- Integrated real-time staff shift notifications, attendance logging, and instant WebSocket notifications for facility announcements.

AI4U Assist – Industry-Specific Conversational AI Personas [App Store] [Google Play] Conversational AI / Cross-Platform

Stack: Flutter, Dart, ChatGPT SDK, Supabase, SQLite, Stripe, Google AdMob, In-App Purchases

- Built customized AI persona assistant tailored for healthcare, legal, and business sectors powered by fine-tuned prompt structures and OpenAI APIs.
- Constructed optimized relational schema on **Supabase** with local **SQLite** caching, ensuring sub-second response recall.
- Monetized through blended model combining subscription access (In-App Purchases), Stripe checkout, and Google AdMob banners.

NR QuikScan – Warehouse Inventory & Logistics Scanner [App Store] [Google Play] Enterprise Logistics

Stack: Flutter, Dart, Hardware Barcode Scanner (Code 39/QR), Isar DB (Offline-First), Node.js

- Developed high-throughput warehouse inventory scanning tool featuring real-time barcode (Code 39) and QR code decoding.
- Architected fully offline-first data model using **Isar database**, guaranteeing zero work disruption in network dead-zones across multi-warehouse facilities.

EDUCATION

Bachelor of Science in Computer Science (BS-CS) 10/2016 – 11/2020

Government College University (GCU), Lahore